# Tachyon® 100G

**Ultra Low Loss Laminate and Prepreg**

**Tg 215°C Td 360°C Dk 3.02 Df 0.0021**

**IPC-4103 /17 IPC-4101 /102  UL - File Number E41625**

## Tachyon 100G laminate materials are designed for very high-speed digital applications up to and beyond data rates of 100 Gb/s.

### PRODUCT FEATURES

Industry Recognition

- UL File Number: E41625
- RoHS Compliant

Performance Attributes

- CAF resistant
- Low moisture absorption
- 6x 260°C reflow capable
- 6x 288°C solder float capable

Processing Advantages

- Multiple lamination cycles
- HDI technology compatible

### PRODUCT AVAILABILITY

Standard Material Offering: Laminate

- 2 to 20 mil (0.05 to 0.51 mm)

Copper Foil Type

- HVLP3 (VLP1) ≤1.1 micron Rz JIS
- HVLP (VLP2) ≤2.5 micron Rz JIS
- Embedded resistor foil

Copper Weight

- ½, 1 and 2 oz (18, 35 and 70 µm) available
- Heavier copper foil available
- Thinner copper foil available

Standard Material Offering: Prepreg

- Tooling of prepreg panels
- Moisture barrier packaging

Glass Fabric Availability

- Low Dk glass
- Square weave glass
- Mechanically spread glass

Tachyon 100G materials exhibit exceptional electrical properties that are very stable over a broad frequency and temperature range between -55°C and +125°C up to 100 GHz. These electrical properties provide designers a scalable solution for next generation designs of backplanes and daughter cards, enabling 10x improvements from 10 Gb/s data rates.

Isola has developed Tachyon 100G with the highest level of thermal performance for high layer count line cards. The very low Z-axis CTE makes it a perfect choice for fine pitch BGA applications.  The material is optimized with the use of spread glass to mitigate skew, improve rise times, reduce jitter, and increase eye width/height and that use ultra smooth HVLP (VLP2) 2um Rz copper that significantly reduces conductor losses.

### PRODUCT ATTRIBUTES




### TYPICAL MARKET APPLICATIONS




### ORDERING INFORMATION:

Contact your local sales representative or contact info@isola-group.com for further information.

**Isola Group**
6565 West Frye RoadChandler, AZ 85226 Phone: 480-893-6527 Fax: 480-893-1409

**Isola Asia Pacific**
(Hong Kong) Ltd.12/F, Kin Sang Commercial Centre, 49 King Yip Street, Kwun Tong, Kowloon, Hong KongPhone: 852-2418-1318 Fax: 852-2418-1533

**Isola GmbH**
Isola Strasse 2 D-52348 Düren, GermanyPhone: 49-2421-8080 Fax: 49-2421-808164



Revision No: H © 2026, Isola Group, All rights reserved.

# Typical Values Table

| Property | | Typical Value | Units | Test Method |
|---|---|---|---|---|
| | | | Metric (English) | IPC-TM-650 (or as noted) |
| Glass Transition Temperature (Tg) by DSC | | 215 | °C | 2.4.25C |
| Glass Transition Temperature (Tg) by DMA | | 230 | °C | 2.4.24.4 |
| Glass Transition Temperature (Tg) by TMA | | 210 | °C | 2.4.24C |
| Decomposition Temperature (Td) by TGA @ 5% weight loss | | 360 | °C | 2.4.24.6 |
| Time to Delaminate by TMA (Copper removed) | A. T260<br>B. T288<br>C. T300 | >60<br>>60<br>>20 | Minutes | 2.4.24.1 |
| Z-Axis CTE | A. Pre-Tg<br>B. Post-Tg<br>C. 50 to 260°C, (Total Expansion) | 45<br>250<br>2.5 | ppm/°C<br>ppm/°C<br>% | 2.4.24C |
| X/Y-Axis CTE | Pre-Tg | 15 | ppm/°C | 2.4.24C |
| Thermal Conductivity | | 0.42 | W/m·K | ASTM E1952 |
| Thermal Stress 10 sec @ 288ºC (550.4ºF) | A. Unetched<br>B. Etched | Pass | Pass Visual | 2.4.13.1 |
| Dk, Permittivity | A. @ 2 GHz<br>B. @ 5 GHz<br>C. @ 10 GHz | 3.04<br>3.02<br>3.02 | — | 2.5.5.5 |
| Df, Loss Tangent | A. @ 2 GHz<br>B. @ 5 GHz<br>C. @ 10 GHz | 0.0021 | — | Bereskin Stripline |
| Volume Resistivity | C-96/35/90 | $1.33x10^{7}$ | MΩ-cm | 2.5.17.1 |
| Surface Resistivity | C-96/35/90 | $1.33x10^{5}$ | MΩ | 2.5.17.1 |
| Dielectric Breakdown | | 60 | kV | 2.5.6B |
| Arc Resistance | | 125 | Seconds | 2.5.1B |
| Electric Strength (Laminate & laminated prepreg) | | 60 (1500) | kV/mm (V/mil) | 2.5.6.2A |
| Comparative Tracking Index (CTI) | | 3 (175 -249) | Class (Volts) | UL 746A<br>ASTM D3638 |
| Peel Strength | A. Low profile and very low profile copper foil<br>B. Low profile and very low profile copper foil ... After thermal stress | 0.79 (4.5)<br><br>0.96 (5.5) | N/mm (lb/inch) | 2.4.8C<br><br>2.4.8.2A |
| Flexural Strength | A. Length direction<br>B. Cross direction | 303 (44.0)<br>283 (41.0) | MPa (kpsi) | 2.4.4B |
| Tensile Strength | A. Length direction<br>B. Cross direction | 207 (30.0)<br>172 (25.0) | MPa (kpsi) | ASTM D3039 |
| Young's Modulus | A. Length direction<br>B. Cross direction | 2,551<br>2,417 | ksi | ASTM D790-15e2 |
| Taylor's Modulus | A. Length direction<br>B. Cross direction | 2,264<br>2,197 | ksi | ASTM D790-15e2 |
| Poisson's Ratio | A. Length direction<br>B. Cross direction | 0.165<br>0.156 | — | ASTM D3039 |
| Moisture Absorption | | 0.1 | % | 2.6.2.1A |
| Flammability (Laminate & laminated prepreg) | | V-0 | Rating | UL 94 |
| Relative Thermal Index (RTI) | | 130 | °C | UL 746 |


Revision No: H © 2026, Isola Group, All rights reserved.

## NOTES

Visit our site http://www.isola-group.com for more details.
Revisions:
A: Initial release - 4/17
B: Corrected moisture uptake value 6/18
C: Corrected units for Flexural and Tensile Strength - 8/18
D: Change MOT to RTI - 5/19
E: Changed VLP2 to HVLP to aligned with common industry terms 4/21
F: Changed TMA Tg to 210C, DSC Tg to 215C and DMA to 230C based on long term data 9/22
G: Added HVLP3, Advanced RTF and 2 ounce copper options, added 20 mil thickness option - 3/24
H: Corrected test method for Df - 3/26

Isola, the Isola logo, Astra, Chronon, GETEK, I-Fill, IsoDesign, IsoStack, I-Speed, I-Tera, Polyclad, Stratus, TerraGreen, and The Base for Innovation are registered trademarks or trademarks of ISOLA USA Corp. in the United States and in other countries. Copyright © 2021 Isola Group. All rights reserved.